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Maternal health and incarceration: advancing pregnancy justice through research

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Abstract

This paper will review the state of the science on maternal health and incarceration. We will provide a historical context on women, pregnancy, and mothers as it relates to mass incarceration, considering both structural racism and reproductive justice. We will discuss existing research that documents care, treatment, and outcomes of individuals who are incarcerated while pregnant or postpartum in the United States. We will discuss the implications of carceral exposure on birthing people and their families. By synthesizing current research and relevant policies, we will identify gaps that will then inform a research agenda for the next decade, including methods and content, to address inequities in and improve maternal and infant outcomes among pregnant and parenting people exposed to incarceration.

Keywords Maternal health, Pregnancy, Postpartum, Incarceration, Prisons, Jails

Background

The incarcerated population in the United States (U.S.) has steadily increased since the 1970s (Carson, 2022; Zeng, 2023). Although incarceration rates slowed during the onset of the COVID-19 pandemic and, in some cases, even declined, rates are now rising again, with little evidence of a sustained downward trend (E. A. Carson et al., 2022). This trend is particularly noticeable among women, whose incarceration rates have surged by more than 500% since 1980, making them the fastest-growing subgroup of the incarcerated population (E. A. Carson et al., 2022). While only four percent of the global female population resides in the U.S., the U.S. accounts

for nearly 27% of the world's incarcerated women (Fair & Walmsley, 2022). Criminal legal scholars have described that the root causes of mass incarceration include state policies driven by structural and systemic racism, changing prosecutorial practices, harsh sentencing laws, the expansion of the private prison industrial complex, and the ripple effects of the so-called War on Drugs (Alexander, 2010; Hinton, 2016; Pfaff, 2017). Such policies and practices have led to heightened surveillance and over-policing, disproportionately affecting people living in poverty, especially Black, Latinx, and Indigenous populations of women who are more likely to be incarcerated than their white counterparts. While often overlooked, mass incarceration and its associated systems of racialized surveillance and control have had a distinct impact on Black pregnant individuals and mothers (M. Goodwin, 2020; D. Roberts, 1997).

Pregnant people are not exempt from mass incarceration, and in fact, it has involved them in numerous and often insidious ways that have had negative repercussions for their and their children's health and wellbeing. Framing the intersection of mass incarceration and pregnancy through the lens of reproductive justice offers important analytic insights as well as strategies for

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reform. Reproductive justice is an analytical framework and social justice movement coined and popularized by Black women in 1994. It examines intersecting systems of power and oppression that shape reproductive experiences, centering on those most marginalized in society and advocating for the liberation of all marginalized people, both inside and outside of jail and prison walls (Loretta Ross & Rickie Solinger, 2017). The framework includes four core tenets: (1) the right to have children, (2) the right to not have children, (3) the right to raise children with dignity and in safe and healthy environments, and (4) the right to bodily and sexual autonomy. Through a variety of ways, mass incarceration disrupts each tenet of reproductive justice (Hayes et al., 2020). Briefly, and respectively, such violations are apparent in, for example, (1) the coerced sterilization of incarcerated women, (2) lack of abortion and contraception access in custody, (3) over-policing of communities of color along with economic divestment in these neighborhoods, causing harmful conditions for raising children, and (4) rampant sexual assault in custody and the inability of incarcerated pregnant people to determine the conditions of their birth (Hayes et al., 2020). In this article, we employ this lens of reproductive justice as we review the current state of research on maternal health care services and outcomes for pregnant and postpartum people in custody in the U.S. Further, we identify priorities for a research agenda for the next ten years.

Research on pregnancy prevalence among incarcerated populations is limited, primarily due to a lack of any federal agency assuming responsibility for national data collection on incarcerated pregnant and postpartum people (Bronson & Sufrin, 2019). Until April 2025 the landmark Pregnancy in Prison Statistics (PIPS) study from 2016–2017 was the only national-scale data on pregnancy prevalence and outcomes in prisons and jails (C. Sufrin et al., 2019; C. Sufrin, Jones, et al., 2020). While the number of women behind bars continues to grow, so too do maternal mortality rates in the U.S., with stark racial disparities (Centers for Disease Control & Prevention, 2024). Incarcerated women, including those who are pregnant, are more likely to have histories of abuse, trauma, and mental health disorders that require access to treatment and healthcare services tailored to their unique needs (Knittel, 2019; C. Sufrin et al., 2015). Additionally, data demonstrates that pregnant people in the criminal legal system often present with predisposing risk factors that can increase the likelihood of pregnancy complications and adverse outcomes, including preterm labor, low birth weight, and stillbirth (American College of Obstetricians and Gynecologists, 2021; Hessami et al., 2023; Knight & Plugge, 2005). Such risk factors include a high prevalence of substance use disorders, chronic medical conditions,

mental health conditions, poverty, homelessness, histories of trauma, and lack of social and family support (Baker, 2019; Cavanagh et al., 2022; Hessami et al., 2023). Pregnant people in custody have elevated health and social needs compared to non-incarcerated pregnant people, needs which are then exacerbated by the conditions of incarceration. Thus, in order to advance maternal health equity and improve outcomes, it is essential to understand how the criminal legal system intersects with the maternal health crisis, birth equity, and reproductive justice.

While the War on Drugs alone does not account for mass incarceration, neither its mass proportion nor its targeted racialization (Pfaff, 2017), the constellation of punitive policies toward substance use and related crime has distinctly impacted women in the U.S., especially Black women. Black pregnant people and mothers, in particular, have faced harsher sentencing and criminalization for their substance use, in part due to the vilification and undervaluation of Black motherhood, amplified by heightened media attention around myths like the “crack baby” symbolizing bad motherhood (Goodwin, 2020). The false narratives of Black drug-addicted mothers causing adverse outcomes to their children continue to have harmful ripple effects decades later, which has gone hand in hand with the “policing of wombs” and parenting at the hands of the criminal legal and child welfare systems (M. Goodwin, 2020; D. E. Roberts, 2022). These exaggerated and scientifically disproven depictions served not only to justify the disproportionate incarceration of Black mothers, but also to obscure the fundamental causes of their adverse maternal and infant health outcomes, such as systemic racism and sexism (A. Carson, 2022; Scott et al., 2019; Zeng, 2023). Systemic racism and sexism persist and encompass systems that perpetuate race and gender-based oppression, like the criminal legal system (Braveman et al., 2022; Homan et al., 2021). Taken together, these intersecting structural disadvantages create distinct challenges for pregnant and postpartum people of color, increasing their vulnerability to contact with the criminal legal system and experiences of reproductive injustice (Hayes et al., 2020; R. J. Shlafer et al., 2019).

Although research on incarcerated pregnant, birthing, and postpartum people is limited, existing data show that care and conditions for them is inconsistent and often substandard, revealing systematic violations of reproductive rights and failing to meet maternal healthcare needs, which we will outline in this paper. The systemic neglect by carceral facilities and systems of oppression that sustain inequities emphasizes the need for research, advocacy, and policy initiatives to address the egregious experiences pregnant people behind bars endure.

The reproductive justice violations faced by incarcerated pregnant people are profound, serving as a backdrop to broader structural issues. In this paper, we shed light on what is known about pregnancy care and the experiences of pregnant people¹ in U.S. prisons and jails while drawing attention to gaps in research and policy to advance pregnancy justice for birthing and postpartum people in custody. We start by discussing pregnancy and postpartum prevalence in the criminal legal system, what is known about pregnancy and postpartum health care and experiences seeking and receiving care in custody, highlight other supports and services that may be available to pregnant and postpartum people behind bars, describe the conditions of confinement, discuss policies related to this population, and wrap-up with future directions to move research on the intersections between pregnancy and incarceration forward.

Pregnancy and postpartum prevalence and outcomes

Until 2025, there were no national data documenting the number of incarcerated pregnant individuals or the outcomes of those pregnancies (Bronson & Sufrin, 2019). During this period, the U.S. Bureau of Justice Statistics (BJS) had not included pregnancy outcomes in their periodic state prison or jail surveys, and the National Center for Health Statistics from the Centers for Disease Control and Prevention (CDC) has not included incarceration status in their pregnancy and birth data tracking. Likewise, the CDC's Pregnancy Risk Assessment Monitoring System, which tracks risk factors for adverse pregnancy outcomes, asks recently pregnant respondents if they or their partner was incarcerated during pregnancy, but does not distinguish maternal incarceration (Testa et al., 2022). Another data gap is maternal deaths; the CDC's various reporting systems for maternal deaths do not report incarceration exposure, and BJS's deaths in custody reports do not report pregnancy status. This systematic omission of pregnancy and incarceration data reflects the trenchant disregard for the existence and wellbeing of pregnant and postpartum individuals in custody.

Several retrospective cohort studies at individual prisons and a systematic review have found incarceration to be associated with adverse outcomes of preterm birth, low birth weight, and stillbirth (Baker, 2019; Bard et al., 2016; Carter Ramirez et al., 2020; Hessami et al., 2023; Howard et al., 2008, 2011; Knight & Plugge, 2005).

Incarcerated pregnant individuals have numerous risk factors for morbidity and mortality during pregnancy and postpartum, including high prevalence of mental health conditions, chronic illness, substance use disorders, limited access to health care, and negative structural determinants of health, including racism (Binswanger et al., 2010; Chambers et al., 2021; Hayes et al., 2020; Karvonen et al., 2023; Logue et al., 2022; Maruschak et al., 2021a, 2021b; Steely Smith et al., 2024). Despite this, no studies or surveillance systems have assessed pregnancy-associated morbidity and mortality among incarcerated people. Furthermore, there are no data on the number of postpartum people who enter custody, let alone systematic data on postpartum metrics.

Until 2025, the closest systematic assessment of pregnancy outcomes in carceral settings came from the Pregnancy in Prison Statistics (PIPS) study (Table 1). From 2016–2017, PIPS prospectively collected monthly pregnancy prevalence and outcomes data from 22 state prison systems, all Federal Bureau of Prisons (FBOP) sites housing females, six jails (including the five largest), and three youth detention systems. In total, there were over 3,000 admissions of pregnant individuals to these facilities, which researchers estimated to be over 58,000 to all U.S. facilities in one year (C. Sufrin et al., 2019; C. Sufrin, Jones, et al., 2020). Among the 1,049 pregnancies that ended in custody, the majority were livebirths; no maternal deaths were reported. While preterm birth rates were overall low at 5%, some states had rates higher than the national rate of 10%. A substantial proportion of pregnant individuals had an opioid use disorder (OUD), which researchers extrapolated to be nearly 8,000 admissions of pregnant people with OUD to state prisons and jails annually (C. Sufrin, Sutherland, et al., 2020a, 2020b). That abortions were less common (5% of pregnancies) than they are in the general population (18% of pregnancies) reflects, in part, barriers to abortion access for incarcerated individuals, as we describe further below (C. Sufrin et al., 2021).

While PIPS data provided crucial benchmarks of data, they were limited to one point in time and have not been repeated; did not collect individual demographic information such as race; and did not include all 50 states or all 3,000+ jails. Furthermore, maternal deaths were only reported up to 6 weeks postpartum and only if the delivery occurred while the person was incarcerated, but not if they entered postpartum. BJS has, since 2018, reported selected annual pregnancy outcomes for FBOP as part of the First Step Act reporting, and in 2024 reported a feasibility study of collecting maternal health data from all prisons and jails (Bureau of Justice Statistics, 2022; Irazola et al., 2024). In the feasibility study, BJS noted that there has been little research or efforts to collect data

¹ We want to acknowledge that people who do not identify as women have the capacity to become pregnant, and thus we use the term 'pregnant people' when referring to this population. However, when describing existing data, we use the term the authors selected when referring to pregnant people which is often 'pregnant women' or 'pregnant females.'

Table 1 Selected PIPS Outcomes, 2016–17

Outcome	Total	State Prisons (n = 22)	FBOP (n = 26)	Jails (n = 6)	Youth Detention (n = 3)
Pregnancy admissions, n(%)	3,089	1,224 (4)	172	1622 (3)	71
Pregnancies ended in custody	1,049	742	75	224	8
Livebirths, n(%)	898 (86)	685 (92)	68 (91)	144 (64)	1 (13)
Abortions, n(%)	47 (4)	9 (1)	2 (3)	33 (15)	3 (38)
Miscarriages, n(%)	93 (9)	42 (6)	4 (5)	43 (18)	4 (50)
Ectopic, n(%)	7 (0.7)	2 (0.3)	1 (1)	4 (2)	0
Stillbirth, n(%)	4 (0.4)	4 (0.5)	0	0	0
Preterm birth, n(%)	51 (5)	39 (6)	NR	12 (8)	0
Opioid use disorder, n(%)	-	117 (26)	NR	50 (14)	NR
Postpartum, ave/mo	-	55	NR	22	NR
Breastfeeding, ave/mo	-	8	NR	6	NR
Psychiatric dx, ave/mo	-	122	NR	43	NR
Hypertension, ave/mo	-	12	NR	10	NR
Diabetes, ave/mo	-	7	NR	5	NR

1. % of all female admissions

2. % of all pregnancies that ended in custody

3. % of all livebirths

4. 6-month outcomes as reported by selected facilities

NR Not reported

Citations: (Asiodu et al., 2021; Hendricks et al., 2024; C. Sufrin et al., 2019, 2021; C. Sufrin, Jones, et al., 2020; C. Sufrin, Sutherland, et al., 2020)

on maternal health in carceral settings, yet it is critical to improving the health of pregnant and postpartum women in custody.

In 2025, BJS released their findings from a one-time survey of maternal healthcare and pregnancy prevalence and outcomes in state prisons and FBOP reporting data from 2023 (Maruschak, 2025). The BJS report collected 2023 data from all 50 states' departments of corrections and the Federal Bureau of Prisons (FBOP). It included statistics on (1) pregnancy testing and positive tests among female admissions; (2) pregnancy prevalence and outcomes by type; (3) pregnancy-related training for staff, emergency transportation protocols, and medical services provided to pregnant and postpartum women; (4) accommodations and support services for pregnant and postpartum women; and (5) provision of and participation in nursery or residential programs where mothers and children reside (Maruschak, 2025). Of the 88% of female admissions tested for pregnancy across 46 jurisdictions, 2% tested positive equating to 1,157 pregnancy admissions to prison in 2023. There were 727 pregnancy outcomes documented between January 1, 2023 and December 31, 2023 including 665 live births, 47 miscarriages, and 15 abortions (Maruschak, 2025). Notably, the data collection did not include maternal deaths in custody, nor did it address women who enter custody postpartum. Although this report provides access to needed

national data for the first time, it has some limitations reflected in part by the inconsistencies with existing data that use real time reporting from state DOC programs.

Nonetheless, gaps in systematic and ongoing national data collection on pregnancy outcomes remain, especially in jail settings, —including pregnancy-associated deaths – among people incarcerated during pregnancy or postpartum. Given that the U.S. has the highest rate of maternal deaths among all high-income countries, and that this maternal mortality crisis affects Black and Indigenous women at disproportionately higher rates, the neglect of maternal health statistics behind bars represents a serious and discriminatory data omission in the U.S.'s overall response to maternal mortality. A further data elision is that, to our knowledge, there has never been a report or assessment of the number of individuals who are postpartum—gave birth within the last 12 months—at time of entry to a carceral facility, or of prevalence of postpartum individuals in custody. To assess these concerns, Congress asked the U.S. Government Accountability Office (U.S. GAO) to provide fact-based, nonpartisan information on issues related to maternal health care in state prisons and jails including (1) what data are available on pregnant women and pregnancy outcomes in state prisons and local jails, (2) what federal assistance can be used to support maternal health care in custody settings, (3) how maternal health care

is provided in select state prisons and local jails, and (4) what are identified and reported challenges and opportunities to providing maternal health care to incarcerated pregnant women (G. Goodwin, 2024). Several of their findings align with this review.

Access to care in custody

The PIPS study demonstrated the prevalence of pregnancy in U.S. prisons and jails and spotlighted the lack of national, state, and local data on this population. The 2023 BJS survey provides additional context on pregnancy in carceral settings. Care in custody varies widely because there is no national requirement that facilities follow set standards or the guidance that has been set by numerous professional organizations regarding the care of pregnant people in custody. Data remain scarce but existing literature begins to illuminate the harsh realities of accessing pregnancy care behind bars, care needs in custody during the perinatal and antenatal periods, and how pregnancy care, or lack thereof, is experienced by pregnant incarcerated individuals.

Screening for pregnancy and postpartum status

Studies of incarcerated pregnant women have shown that many of them first learn about their pregnancies upon arrival to prison or jail (C. Sufrin, 2017; C. Sufrin et al., 2024; C. B. Sufrin et al., 2023). National guidelines recommend offering all pregnant-capable people younger than 55 a pregnancy test at intake to a carceral facility (American College of Obstetricians and Gynecologists, 2021). Yet studies of intake and other screening protocols at carceral facilities demonstrate variability in whether and when pregnancy tests are administered. For instance, in the PIPS study, 4 of 22 prisons only performed urine pregnancy testing at the discretion of a clinician or request of patient, but did not routinely screen for pregnancy status, while 14 (64%) of prisons routinely tested at intake (C. Kramer et al., 2023). A national survey of over 800 jails found that only 31% routinely performed pregnancy tests at intake or within two weeks of arrival (C. Sufrin et al., 2022). The 47 jurisdictions (state DOCs and FBOP) that reported pregnancy testing in the 2023 to BJS noted that 88% of female admissions were tested upon arrival (Maruschak, 2025). Without systematic, routine pregnancy screening practices, carceral facilities—and patients—are unaware that there are pregnant patients in their facility, which then makes it feasible for facilities to claim that improvements in and expansion of pregnancy care would not be necessary. Thus, this lack of standard pregnancy screening further perpetuates systemic deficiencies in care. We are not aware of any studies that report the frequency of screening for postpartum status

at entry, which reflects the lack of data on prevalence of postpartum individuals in custody.

Prenatal care in custody

Pregnancy is a significant physical and mental transitional period that requires time-sensitive specialized care to ensure a healthy pregnant individual and fetus. Pregnant incarcerated people have numerous social and structural determinants of health, like racism, that predispose them to a higher risk of maternal morbidity and mortality, including a higher prevalence of chronic disease, mental health diagnoses, and substance use disorders (Baker, 2019; Cavanagh et al., 2022; Chambers et al., 2021; Hessami et al., 2023; Karvonen et al., 2023). Pregnant incarcerated individuals often have lower rates of prenatal care utilization in the community and experience chronic food insecurity due to the overrepresentation of socioeconomically marginalized populations in jails and prisons (Baker, 2019). Prenatal care services in carceral facilities cannot mitigate all these factors, but it means that it is even more essential for them to provide access to quality, comprehensive prenatal care. Data show that this is not consistently the case, as outlined below. What's more, the carceral environment by nature is punitive, coercive, and rights-removing and is the antithesis of a healthy environment for fetal development. Pregnancy-related care in prisons and jails is variable and often far below community standards of care. At best, carceral pregnancy care is accessible but is often inadequate, and sometimes non-existent (Hessami et al., 2023). Care deficiencies may exist because women make up only a fraction (~ 10%) of incarcerated individuals (Carson & Kluckow, 2023; Zeng, 2023) and because carceral institutions were designed for and are still predominantly run by men. The male-centric setting is evident by the limited facility policies and procedures related to women's health and pregnancy care (C. Kramer et al., 2023).

In 2004, a survey of incarcerated people by BJS reported nearly 94% of pregnant individuals in state prisons received an obstetric exam and a little more than half (54%) received prenatal care (Maruschak, 2008). The prior report from 1997 did not collect any data on pregnancy or obstetric care (Maruschak & Beck, 2001). Until 2025, the most recent survey of incarcerated people from BJS was collected in 2016 and documented that 91% of pregnant females in state prisons and 87% in federal prisons received an obstetric exam (Maruschak et al., 2021b). Only about half of the pregnant females in state (50%) and federal (46%) prisons received some form of prenatal care such as instructions on childcare, exercises, a special diet, medication, or special testing from a health care provider. In the 12-year difference between the BJS surveys, prenatal care access for incarcerated pregnant

people in prison worsened. BJS reported that 96% of surveyed prisons provided a medical appointment within two weeks of a known pregnancy in 2023 (Maruschak, 2025).

Structural barriers within the carceral environment create challenges to pregnancy care delivery, like limited travel to off-site care appointments. The PIPS study assessed pregnancy care policies in carceral facilities and found that 55% of study prisons and 67% of study jails delivered routine prenatal care through a combination of on-site and off-site arrangements (C. Kramer et al., 2023). In the 2023 survey, BJS reported that all state prisons and FBOP had an infrastructure for caring for pregnant women either via their on-site infirmary (96%) or through 24/7 or on call care (98%) (Maruschak, 2025). A survey of 52 carceral facilities (96% jails) across 40 U.S. states found that 31% of study jails did not have on-site OB/GYN care with jail staff respondents reporting they have to “jump through hoops” to get women proper care off-site which was described as hurting “convenience, cost, and security” (Kelsey et al., 2017). Some facilities require the patient to pay a co-pay for pregnancy care. That same survey of jails conducted in 2014–2015 demonstrated that 86% of jails do not charge for OB/GYN services but those that did charged an average of \$15 per visit (range = \$0–\$35), creating financial barriers for patients (Kelsey et al., 2017). A qualitative interview study with 34 jail staff across five southeastern states described how workers perceived pregnant incarcerated people as a liability with limited available resources to tend to their needs which resulted in viewing pregnant people in custody as a burden to staff and the facility (A. K. Knittel et al., 2023). Jail staff attitudes were characterized by stigma associated with incarceration and substance use, distrust of pregnant incarcerated women, and a heightened focus on fetal safety. Furthermore, there is little oversight and monitoring of carceral health care services, in general, which leaves decisions of what care should be delivered and how it is delivered to carceral staff.

A systematic review from 2022 that compared prenatal care for incarcerated pregnant women ($N = 2,544$) to non-incarcerated pregnant women ($N = 8,990$) demonstrated that deficiencies in carceral prenatal care persist (Hessami et al., 2023). The study used a matched control group to assess adequacy of pregnancy care behind bars and found that 34% of pregnant incarcerated women had inadequate prenatal care and were at a higher risk of adverse birth outcomes like low birthweight compared to non-incarcerated pregnant individuals. Another study that examined the medical records of 147 infants born to women who delivered at-term while incarcerated in a Texas prison found a statistically significant increase for infant birthweight with each additional prison prenatal

care visit demonstrating the impact of prenatal care on fetal development and health (Howard et al., 2008). This was only among study infants whose mothers entered prison during the first trimester.

Several studies on health care for women behind bars, including qualitative accounts from patients, report multiple barriers to accessing care including punishment for seeking care, long wait times resulting in delayed or absent care, a lack of response from medical providers, and health concerns being ignored (Cavanagh et al., 2022; C. Sufrin et al., 2024; C. B. Sufrin et al., 2023; Wennerstrom et al., 2022). A survey of 192 jail administrators in the midwestern U.S. from 2017–2018, found that jails most often test for pregnancy (60%) and distribute prenatal vitamins (44%) (Lipnicky et al., 2023). These proportions are relatively low for services that are inexpensive and require little time. Jails offering pregnancy services were significantly more likely to be in urban areas and had more general physicians, mental health providers, and mid-level providers. Data from the 2023 BJS prison survey indicated higher rates of pregnancy testing (88%) and providing prenatal vitamins (100%) (Maruschak, 2025).

Pregnancy care in custody varies by facility based on several factors, but qualitative research suggests that people who are incarcerated experience more discrimination and derogatory treatment for the simple fact that they are incarcerated (C. Sufrin, 2017; C. Sufrin et al., 2024). As noted, discrimination and systemic racism in the U.S. is associated with disproportionately higher rates of maternal mortality among Black Americans, who are also disproportionately incarcerated, regardless of other characteristics like socioeconomic status and education levels (A. Knittel & Sufrin, 2020). And, the number of prenatal care visits alone has not been sufficient in reducing maternal mortality, thus increasing our understanding that access to prenatal care services in prisons and jails are essential but the quality and experience of care cannot be overlooked.

Maternal mental health and substance use disorders in custody

The leading cause of pregnancy-related deaths in the U.S. is mental health conditions (Troost et al., 2022). Given the high prevalence of mental health conditions among incarcerated pregnant and non-pregnant women, including substance use disorders (Bronson et al., 2017; Hendricks et al., 2024; Maruschak et al., 2021a), this raises concern for the high risk of maternal mental health morbidity and mortality for pregnant and postpartum people during and after incarceration. Many studies document dehumanizing conditions for pregnant incarcerated people characterized by inadequate prenatal care, the trauma

of being shackled during pregnancy and childbirth, isolation during delivery, invasions of privacy, the stigma of being pregnant while incarcerated, and the lack of mental health support (Kirubarajan et al., 2022). This dehumanization then worsens existing mental health conditions and leads to incarceration-induced stress for pregnant incarcerated people.

According to the 2016 BJS survey of prison residents, females in state (19%) and federal (17%) prisons were more likely to report meeting criteria for serious psychological distress in the last 30 days compared to males in state (14%) and federal (7%) prisons (Maruschak et al., 2021a). Females in state (69%) and federal (52%) prisons were also more likely to have a history of mental health problems compared to males in state (41%) and federal (21%) prisons. The gender disparities in mental health may be attributable to the extremely high rates of sexual, physical, and emotional trauma women experience before incarceration, an astounding 90% reported in one study (Lynch et al., 2012), that leads to lifelong mental health conditions. Despite the documented prevalence of mental health conditions among women behind bars and the expressed need for care, mental health treatment in prisons and jails is lacking. According to the National Alliance of Mental Illness (NAMI), about three in five people with a history of mental illness do not receive mental health treatment while incarcerated, and nearly 50% of individuals on medications for a mental health condition are discontinued at admission to a carceral facility (“Mental Health Treatment While Incarcerated,” n.d.). The PIPS study reported monthly data on the pregnant incarcerated people in custody over a six-month period. They documented that mental health conditions were the most frequently reported chronic medical condition of the newly admitted 445 pregnant people to prisons and the 243 admitted to jails (Hendricks et al., 2024). Substance use was the most prevalent in prisons (34%) and jails (24%) followed by a psychiatric diagnosis (27% and 18%, respectively). Eighty-eight percent of the 45 jurisdictions that responded in the 2023 BJS survey reported providing pregnant and postpartum women in prison with access to a psychologist (Maruschak, 2025), though there is little known about what that care looks like.

Fatal overdoses among pregnant and postpartum people increased by 81% from 2017 to 2020 (Bruzeliuss & Martins, 2022), and the CDC found that mental health conditions, including substance use disorders (SUDs), accounted for 23% of pregnancy-related deaths in 2022 (CDC, 2024b). Pregnant people with SUDs, including OUD, are particularly vulnerable to stigma, discrimination, mental health conditions, and criminal legal involvement due to the criminalization of substance use in pregnancy (Murphy, 1999; Weber et al., 2021). The

American College of Obstetricians and Gynecologists (ACOG) and other leading medical organizations oppose the criminalization of drug use during pregnancy because evidence demonstrates punitive actions are ineffective at changing behaviors, are harmful to pregnant people and their infants, and deter pregnant individuals from seeking prenatal care and SUD treatment (*Opposition to Criminalization of Individuals During Pregnancy and the Postpartum Period*, n.d.). Furthermore, research consistently shows that punitive measures do not prevent substance use, which is now recognized as a medical condition that requires diagnosis and treatment (*More Imprisonment Does Not Reduce State Drug Problems*, 2018; Volkow, 2021). Despite this, 25 states and the District of Columbia have laws that consider substance use during pregnancy to be child abuse or neglect (Faherty et al., 2019), yet only seven states have laws directly related to SUD screening and treatment among pregnant incarcerated populations (Steely Smith et al., 2023). Policy makers may argue that punitive state laws prevent pregnant people from using drugs, but one study found significantly greater rates of neonatal abstinence syndrome (NAS) across eight states in the first year after such state laws were passed (Faherty et al., 2019).

Pregnant incarcerated individuals who use opioids frequently experience forced withdrawal in custody, despite the longstanding evidence and recommendation against opioid withdrawal in pregnancy. Medications for opioid use disorder (MOUD) in custody for pregnant people remains insufficient putting the estimated 8,000 pregnant people with OUD in prisons and jails annually at risk for adverse health outcomes (A. K. Knittel et al., 2020; C. Sufrin et al., 2022; C. Sufrin, Sutherland, et al., 2020a, 2020b). A jail policy review study documented that some jails lack policies and protocols on the provision of MOUD for pregnant people in custody (Kao et al., 2023). The policy review also noted that jail policies contradicted their reported practices when caring for pregnant people with OUD in custody. A qualitative study with pregnant and postpartum women who were in jail while pregnant with OUD described withdrawal in harrowing terms; even when MOUD was provided in jail it was traumatic and characterized by stigma, discrimination, and inaccurate medical information (Lingerfelt et al., 2024; C. Sufrin et al., 2024). Furthermore, a national survey of jails, with 836 responses, on MOUD availability for pregnant and postpartum people in custody, found that nearly 76% of jails that provided MOUD in pregnancy discontinued treatment postpartum if the individual remained in custody (C. Sufrin et al., 2022). This is particularly dangerous given that most pregnancy-related opioid deaths occur postpartum and that the risk

of overdose is quadrupled when pregnant people with OUD are incarcerated (Nielsen et al., 2020).

Postpartum people in custody experience higher rates of postpartum depression (PPD) (35–80%) compared to the general population (12–20%) (Chawla et al., 2024). Risks for postpartum depression include a previous mental health diagnosis, lack of social support, poor perinatal care, inability to breastfeed, lack of skin-to-skin contact, and partner violence, all of which depict the experiences of pregnant incarcerated people. Immediate child separation after birthing in custody can lead to profound loss and grief for mothers in an environment with little, if any, emotional support (Ferszt & Clarke, 2012). In a qualitative study of 17 pregnant people in a state prison, participants expressed the anticipated trauma of birthing in custody and impending child separation that led to the fear of being postpartum in prison due to the lack of mental health support (C. T. Kramer et al., 2025). PPD, if untreated, can lead to a poor quality of life for the mom and infant and manifests as sleep disorders, mood swings, appetite changes, social isolation, anhedonia, and suicidal thoughts/ideation (Bigelow et al., 2012; Ghaedrahmati et al., 2017). Women who are incarcerated for longer periods after birth and child separation report higher levels of postpartum depressive symptoms (Howland et al., 2021). Validated screening tools like the Edinburgh Postnatal Depression Scale should be integrated into standard carceral medical practices both for people who give birth in custody and for people who enter custody postpartum, which is often not assessed at intake. A quality improvement study at the Milwaukee County Jail on perinatal depression screening was found to be effective at standardizing screening and treatment and is replicable across carceral facilities to improve equitable care (Meine, 2018). BJS reported that 47 prisons (92%) provided depression screening during pregnancy and postpartum in 2023 (Maruschak, 2025).

Abortion access, miscarriage care, and the implications of *Dobbs*

In 2021, the CDC reported nearly 88% of all abortions were for women in their twenties (57%) and thirties (31%); 50% were among women of color with Black women comprising 42%; and 61% of abortion patients already had a least one child (Kortsmit, 2023; Ranji et al., 2024). Abortion rates in the U.S. follow similar disparities with people of color, those with lower income and less education, and those who experience more adverse health outcomes, including infant mortality, receiving abortions at higher rates (Dehlendorf et al., 2013). Black and other women of color of reproductive age who are caregivers to young children and experience poverty and systemic inequalities are also more likely to be

incarcerated. The denial of abortion has long-term health and other consequences for women that have been documented by the revolutionary prospective longitudinal Turnaway Study that examined the effects of unwanted pregnancy on women's lives (*The Turnaway Study*, n.d.). They found that being denied an abortion is harmful and results in economic hardship, insecurity, and worse health and family outcomes. Abortion access and experiences seeking and obtaining an abortion in custody are understudied.

Before the Supreme Court overturned *Roe v. Wade* (Roe) in June 2022, all pregnant incarcerated people had a constitutional right to abortion in custody. However, the PIPS study (2016–2017) documented that of the 22 state prison systems, three facilities did not allow abortion at all with no exceptions for rape or incest, and seven prisons did not have a written policy about abortion (C. Sufrin et al., 2021). Of the six study jails, two did not allow abortion with one jail's policy explicitly prohibiting the procedure. PIPS documented a low rate of abortion (5% of pregnancies) (C. Sufrin et al., 2021) which fell below the general population abortion rate (18%) during that time (Jones et al., 2019). Fewer than expected abortions may occur in custody due to factors like little to no logistical assistance in arranging, paying for and getting an abortion, the lack of official facility policies for abortion, and custody providers' denying, delaying, or ignoring requests for abortions (Kasdan, 2009; C. Sufrin et al., 2021; C. B. Sufrin et al., 2023). One study used qualitative interviews with pregnant incarcerated women ($N = 39$) in two state prisons and two jails to assess abortion access, options counseling, and how incarceration affected their thoughts about their pregnancy, birth, abortion, and parenting (C. B. Sufrin et al., 2023). No one in this study received an abortion in custody but participants shared that the medical providers overtly obstructed desired abortions; some assumed they didn't have a right to an abortion due to being incarcerated; carceral bureaucracy constrained abortion access; and carceral conditions made women wish they had aborted. There are no current studies on the experiences of pregnant people who were able to obtain an abortion in custody.

Access to legal abortion in the U.S. is severely restricted or banned since the Supreme Court's *Dobbs v. Jackson Women's Health Organization* (*Dobbs*) decision (*State Bans on Abortion Throughout Pregnancy* | Guttmacher Institute, 2024). We know from the literature that carceral facilities denied abortion access, before *Dobbs*, despite the absence of any actual, or perceived, legal barriers (Kasdan, 2009). One team started to track the repercussions of *Dobbs* on pregnancy care through their Care Post-Roe study by documenting cases of poor-quality care for pregnant people in community settings (*Care*

Post-Roe, n.d.). By collecting stories from medical professionals, their preliminary findings demonstrate that the fall of *Roe* led to treatment delays, denials in care, worsened health outcomes, and an increased financial and emotional burden on patients. They conclude that abortion bans are essentially eroding health care for pregnant people in the U.S. According to BJS' 2023 data published in 2025, only 2% of prison pregnancies ended in abortion with no data on abortion requests in restrictive states (Maruschak, 2025). There are no current data on how *Dobbs* is impacting pregnant incarcerated people who have limited autonomy and, who unlike people in the community, do not have freedom to travel to another state for care. Pregnant people on parole and probation in the community may be under similar constraints and may need court approval to seek an abortion in another state.

There is even less data available on pregnancy loss behind bars. PIPS documented that there were 46 miscarriages (6%) of the 1396 pregnant women in prison and 41 miscarriages (18%) of the 224 jail pregnancies (C. Sufrin et al., 2019; C. Sufrin, Jones, et al., 2020). Six percent of prison pregnancies in 2023 ended in miscarriage (Maruschak, 2025). Miscarriage in the U.S. is common with 43% of women reporting having had one or more first trimester miscarriages (Cohain et al., 2017). About 2–3% of pregnancy loss occurs in the second trimester. All miscarriages necessitate medical diagnosis and treatment, but second trimester miscarriages require a higher level of care to ensure the safety of the mother (*Understanding Second Trimester Miscarriage* | Family Planning | Obstetrics and Gynecology | UC Davis Health, n.d.). The Care Post-Roe study received many narratives of how care for inevitable miscarriages is being criminalized due to *Dobbs* putting the mother's life at risk for poor health outcomes including death (*Care Post-Roe*, n.d.). If miscarriage care in the community has been impacted, we can infer similar outcomes for pregnant incarcerated people.

Labor, birth, and infant placement in custody

During the PIPS study, 897 of the 3,018 pregnancies (30%) ended with livebirths in custody (C. Sufrin et al., 2019; C. Sufrin, Jones, et al., 2020). Birthing and infant visitation policies and practices were also reported by PIPS prisons and jails. Most facilities reported stationing an officer inside the hospital room during labor and delivery with one-third of them not requiring a female-identifying officer (C. Kramer et al., 2023). This practice directly violates the Prison Rape Elimination Act (PREA) standards of restricting cross-gender viewing of an incarcerated person's exposed buttocks, breasts, or genitalia (United States Department of Justice, n.d.). It is also an invasion

of privacy by allowing a stranger to witness childbirth, one of the most vulnerable and intimate moments, when the risk for escape or harm is minuscule due to being in active labor. Less than half of the PIPS sites allowed visitors, who may act in a supportive role, during childbirth or postpartum hospitalization (C. Kramer et al., 2023). Infant contact policies after birth varied by facility with a few providing special accommodations to postpartum people, most only allowing regular visiting hours, and some not allowing any form of contact with the newborn. ACOG recommends that incarcerated postpartum people be allowed the maximum time for parent-infant bonding during birth hospitalization to avoid detrimental effects on parent-infant bonding, breastfeeding, and psychological wellbeing ("Reproductive Health Care for Incarcerated Pregnant, Postpartum, and Nonpregnant Individuals," 2021). However, in practice, the length of time an incarcerated postpartum mother can spend with her infant in the hospital after birth varies but could be as short as 24 h as prison and jail staff are eager to not have their residents and custody staff off-site for such an extended period (Franco et al., 2020). Only eight states have laws that allow new mothers to remain with their infants for at least 72 h after birthing in custody (Moss et al., 2024). Of the 727 pregnancies that ended in prison in 2023, 665 of them were live births (Maruschak, 2025). BJS also reported that 35% of the prisons provided doula services, but it did not provide details on infant visitation.

According to ACOG, labor inductions, the medical stimulation of contractions in the uterus, are necessary if the life of the mother or fetus is at risk for adverse health outcomes (*Labor Induction*, n.d.). Elective inductions occur when one chooses to induce labor and should only be done if the mother and fetus are healthy and the pregnancy is at least 39 weeks gestation. Labor inductions should not be routine practice, especially in carceral settings where prenatal care is often inadequate and many pregnancies are considered high-risk. One study reviewed hospital electronic medical records to examine labor and delivery characteristics of 219 women who received prenatal care in custody between 2014–2019; 146 pregnancies ended in a live birth in custody (Steely Smith et al., 2024). They found that over a third ($N=49$) of those who gave birth in custody were induced, most of them being post-term pregnancies (i.e. >40 weeks gestation); six inductions were reported as elective, and one was pre-term (i.e. <37 weeks gestation). The practice of using labor inductions to schedule birth behind bars for carceral convenience is underexplored and warrants more research to ensure pregnant people in custody who are already at a high-risk for adverse outcomes due to predisposing conditions and exposure to the

criminal legal system are not being coerced and forced to birth prematurely.

Regardless of what care is available, pregnant incarcerated people have no control over the timing of care which is dictated by carceral policies and practices. Thirty-nine pregnant incarcerated individuals shared in qualitative interviews that the unknown for patients creates profound fear and anxiety as pregnancies progress and births near (C. B. Sufrin et al., 2023). Follow-up interviews with a subset of these participants during the postpartum period described the trauma of birthing in custody without sufficient postpartum support and how infant separation negatively affected their mother-infant bonds (C. T. Kramer et al., 2025).

Despite national and international medical standards that advise against the use of restraints while pregnant because they increase the risk of falls, placental abruption, and other injuries, and interfere with the need for emergent medical procedures, among other risks (American College of Obstetricians and Gynecologists, 2021), research documents that shackling still occurs regularly, even in states with anti-shackling legislation, which we discuss further below (Kramer et al., 2023). Some reasons for these deviations in the standard of care are that hospital staff often are unaware of state laws or best practice and stigma and judgement may lead to mistreatment. For instance, a cross-sectional study of 923 perinatal nurses reported that 83% of respondents had an incarcerated patient who was shackled at some point in the hospital perinatal unit despite only 10% reporting they ever felt unsafe with an incarcerated pregnant woman (L. S. Goshin et al., 2019). Alarming, only 3% of the nurses correctly identified the conditions under which shackling may be ethical. The same survey of nurses identified that lower individual and institutional stigma was associated with higher care intentions (L. S. Goshin et al., 2020). Additionally, state laws have loopholes that allow custody officers and hospital personnel to over-apply exceptions (DiNardo, 2018).

If the birthing person remains in custody postpartum, she is tasked with making arrangements for who will care for the baby after birth. Infant placement arrangements and navigating the child welfare system, if necessary, can be difficult to make behind bars when phone or internet access is restricted. In a qualitative study with 17 pregnant incarcerated individuals, women expressed extreme anxiety in general over infant placement that was heightened by the lack of support from prison staff and little information about how to navigate the process (C. T. Kramer et al., 2025). Some participants' main concerns were keeping their babies out of the foster care system, yet they felt like they had little control over the process. According to PIPS, most infants (68% in prisons and 74%

in jails) are placed with a designated family member (Asiodu et al., 2021). However, 10% of infants born to mothers in prison and 5% in jails were placed in foster care. There are very few studies that assess infant placement procedures and nuances for birthing people in custody. While mother-infant care programs, also known as nursery programs, exist and allow people who give birth in custody to bring their babies back to prison with them, data on the benefits are mixed, as we describe below.

Postpartum care and breastfeeding in custody

Postpartum care in custody is critical given the documented number of pregnant people who give birth in prisons and jails each year. Nearly two-thirds of pregnancy-related deaths in the U.S. occur in the postpartum period (*Insights into the U.S. Maternal Mortality Crisis*, 2024). Postpartum people require timely and routine check-up appointments in the 12 weeks after birth to screen for and address complications that are specific to the recently pregnant and postpartum status that can be harbingers of severe maternal morbidity or mortality (ACOG, 2018). People who undergo a C-section in custody, which is major abdominal surgery, require more medical oversight to prevent postoperative complications, but such oversight or ideal wound healing conditions are not always present in custody. Moreover, urgent maternal warning signs that can indicate underlying post-pregnancy complications can occur up to 12 months after birth while the body is returning to its pre-pregnancy state (ACOG, 2018), yet it is unlikely that carceral facilities recognize this fact and its attendant vigilance for postpartum symptoms. Importantly, there is a dearth of data on whether carceral facilities screen for postpartum status at intake. Since many facilities still do not consistently screen for pregnancy, we can assume postpartum people are not identified upon incarceration and may not get the healthcare they need.

Breastfeeding is recommended due to numerous well-established maternal and infant health benefits in the short and long term, such as reducing infant infections and reducing postpartum individual's risk of cancer, diabetes and high blood pressure (CDC, 2024a). Breastfeeding is a public health issue with stark disparities rooted in slavery, racism, poverty, the objectification of female bodies, and the ideology of good motherhood (Beit, 2022; Kirksey, 2021). The controversial history of breastfeeding in the U.S. has underpinnings that derived from wet nursing both through the exploitation of enslaved Black women who were forced to breastfeed white babies instead of their own children and through monetary exchange for poor women; the stigmatization of breasts, particularly Black female bodies; the insufficient support from the medical establishment to promote

breastfeeding; and state surveillance over breastfeeding for mothers who received support from the Women, Infant and Children's (WIC) program, primarily Black women and those living in poverty.

According to ACOG, postpartum individuals in custody should receive breastfeeding education in addition to lactation support and accommodations to express and store breastmilk for their infants ("Reproductive Health Care for Incarcerated Pregnant, Postpartum, and Non-pregnant Individuals," 2021). The PIPS study documented that 16 of the 28 carceral study facilities (11 prisons and 5 jails) had written policies that supported pumping or breastfeeding in custody (Asiodu et al., 2021). Carceral policies on breastfeeding vary from nothing to full accommodations. Accommodations included allowing mothers to directly breastfeed their infants during contact visits, only allowing pumping and storing the milk for the infant, and only permitting "pumping and dumping" so women can maintain their milk supply (Asiodu et al., 2021). About 78% of prisons in BJS' 2023 survey allowed postpartum women to pump breastmilk, however it is unclear whether there were allowable arrangements to get the milk to the infant (Maruschak, 2025). Maternal incarceration directly disrupts the ability to breastfeed, with rare exceptions during infrequent infant contact visits and at the few sites with mother-infant care programs. Only six states have infant feeding policies for incarcerated pregnant people (Moss et al., 2024).

Qualitative studies with pregnant incarcerated women demonstrate their desire to breastfeed. Twenty pregnant women at the New York City jail indicated that being incarcerated created uncertainty around breastfeeding but being in custody, away from their communities, gave them a new start to motherhood, one in which they wanted to breastfeed (Huang et al., 2012). Another study of 17 pregnant incarcerated people in a state prison found that women desired to breastfeed and believed it would help facilitate the mother-infant bond (C. T. Kramer et al., 2025). A mixed-methods evaluation study of a prison pregnancy program demonstrated that incarcerated pregnant women who discussed breastfeeding with their prison-based doula were seven times more likely to initiate breastfeeding than those who did not (R. J. Shlafer et al., 2018). Qualitative excerpts from doulas also identified barriers to breastfeeding in custody including infant separation and lack of encouragement from hospital staff likely due to stigma.

Pregnancy in youth detention facilities

The 2004 Juvenile Facilities Census reported that 2.1% of girls are pregnant in juvenile facilities at any given time (Committee on Adolescence et al., 2011). The PIPS study assessed pregnancy prevalence and outcomes

among three juvenile residential facilities documenting 71 admissions of pregnant adolescents over a 12-month period equating to 3.3% of female population in juvenile facilities (Kim et al., 2021) (Table 1). Eight of those pregnancies ended in custody with one live birth, four miscarriages, and three induced abortions. PIPS data indicated that most pregnant teens return to the community while still pregnant highlighting the increased need for adequate care linkages at release. The 2004 census survey noted challenges in caring for pregnant adolescents in custody given that nearly 25% of juvenile facilities do not offer access to obstetric services (Committee on Adolescence et al., 2011). Pregnant youth in detention face similar disparities to pregnant women in the criminal legal system like sexism, racism, and maternal health inequities. More research is needed to understand and address the unique treatment barriers and experiences of pregnant youth in detention.

Other services and supports

Some prisons and jails have implemented specific programs to begin to address the needs of pregnant people in custody. Bard and colleagues (2016) reviewed fifteen studies that examined various models of perinatal health care within carceral facilities, ranging from routine pregnancy care in these facilities to additional support, such as lactation support and doula care. They found that while there is limited data on these programs, they improve both short and long-term outcomes. Enhanced pregnancy programs in the review indicated increased access to prenatal vitamins and iron supplements, nutritious food, prenatal counseling and education, reduced physical duties, and, in some instances, pregnant people were transferred to separate facilities during the third trimester (Bard et al., 2016). Their connections to community-based health services are an attempt to bridge care between carceral facilities, the community, and public health systems to improve carceral conditions and pregnancy care.

Pregnant people who are incarcerated have limited control over their diets and often rely on the facility to provide the recommended dietary intake for prenatal nutrition to ensure a healthy pregnancy (R. Shlafer et al., 2017). A study of William and Mary Healthy Beginnings Program (W&M HBP), which provided prenatal vitamins, pregnancy tests, nutrition education, and reentry support to residents in seven jails in a southeastern state compared birth outcomes to changes in nutrition and pregnancy knowledge among two comparison groups (Dallaire et al., 2017). They found that those who participated in the W&M HBP had better birth outcomes (i.e., higher birth weights and longer gestational lengths)

than the comparison group who did not participate. The researchers also found a significant improvement in nutrition and pregnancy-related knowledge after participants received counseling from a W&M HBP team member signifying that intentional pregnancy programming and care improves birth outcomes in custody.

Some carceral facilities provide access to doulas, trained labor support persons, to pregnant people in custody. Doula support for incarcerated people varies, but it can include visits to review labor and birth expectations, continuous support during labor and birth, and postpartum support to review the birth events (Schroeder & Bell, 2005). Doula support has been growing in both the community and carceral settings due to its positive impact on birth outcomes, particularly in underserved communities (Bohren et al., 2017). Currently, around 20 programs across 16 U.S. states offer doula care to incarcerated pregnant and postpartum people through small group sessions and one-on-one support (Bradley, 2024). These programs offer childbirth education and physical and mental assistance before, during, and after birth. A study of jail officers, medical providers, and pregnant incarcerated women who received doula care in custody revealed that providers ($N=32$) and COs ($N=8$) rated the doula program highly and reported satisfaction on behalf of patients (Schroeder & Bell, 2005). Interviews with the pregnant women who received doula care ($N=17$) demonstrated doulas were extremely important to their birth experiences, helped ease the stress of birthing in custody, and that pictures and stories from doulas of their infants aided with the loss and grief women felt after being separated at birth. Research from the Minnesota Doula Project concluded that doulas provide structure and routine in the carceral setting, which is often limited in environments that do not standardize maternal care (R. Shlafer et al., 2021; R. J. Shlafer et al., 2015). Shlafer et al. (2015) conducted a qualitative study of six doulas at a mid-western state prison, reporting that doulas were able to empower their clients, establish a trusting relationship, normalize delivery, and support women postpartum and post-separation. Prison doulas in this study facilitated initial chest feeding and offered postpartum emotional support even though many prisons restrict chest feeding after a parent is separated from their infant.

Several studies and reports have described programs that allow pregnant and postpartum people to co-reside with their babies in custody after childbirth (Barkauskas et al., 2002; Carlson, 2001; Carlson Jr., 2009; L. Goshin et al., 2014; Tuxhorn, 2022). At yearend 2023, 10 states and the FBOP reported operating a prison nursery program with 86 women participating in such programs (Maruschak, 2025). Although controversial, advocates of prison nursery programs—also known as mother-infant

care programs—argue that they mitigate the traumatizing effects of forced mother and baby separation (Craig, 2009). For example, prison nursery programs, such as those in New York's Westchester County and Ohio's Achieving Baby Care Success program, enable incarcerated mothers convicted of low-level offenses to live with their babies in custody in separate buildings away from the general population. These programs offer parenting classes, maternal and infant healthcare, Head Start readiness programs for toddlers, and coordinated community resources upon release (Byrne, 2019; Carlson, 2018). Due to the limited number of prison nursery programs and their recent inception, there is a paucity of research on their effectiveness. An exploratory comparison study assessed attachment and mother nurturing outcomes among mother-infant dyads for individuals who participated in an out-of-prison nursery program in Texas versus individuals who participated in an in-prison nursery program in Nebraska (Kwarteng-Amaning et al., 2019). They reported no significant differences. However, there are currently no studies that compare attachment and nurturing among prison nursery program participants to those in the community who the criminal legal system has not touched. Existing research indicates that prison nurseries are very selective in who they admit, with strict eligibility criteria like low-level offenses and short sentences (C. T. Kramer et al., 2025; Tuxhorn, 2022). Thus, it can be argued that these programs omit the people who need them the most, the birthing people who remain in custody for longer periods. It is also plausible that the postpartum people who meet these criteria may be able to successfully parent under community supervision rather than in custody. Thus, prison nursery programs should be critically examined.

Conditions and impact of incarceration

Subpar living conditions in custody compound pregnant and postpartum people's experiences behind bars. Studies reveal limited access to nutritious food, prenatal vitamins, optimal sleeping arrangements, clean water, and controlled recreational time (Alirezai & Roudsari, 2022). In more extreme cases, as was the case during the COVID-19 pandemic, pregnant people were placed in solitary confinement, which is condemned in pregnancy, exposing them to severe psychological distress and hindering their ability to access maternal healthcare services (Marsh et al., 2024).

A growing body of research has described the pregnancy and birth experiences of people in custody. In their qualitative systematic review of the literature, Kirubarajan and colleagues (2022) identified 24 studies that captured the perspectives of female patients, healthcare providers, and prison staff. Across the existing literature,

patients reported considerable concerns regarding access to timely and quality mental and physical health care; patients experienced dehumanizing care, limited to no privacy, social isolation and lack of support, and harsh treatment (e.g., use of shackles during labor and birth). Numerous studies in their review described the intense emotional distress caused by the separation of mothers from their newborns. Notably, there is very little research on the experiences of alternative caregivers who take responsibility for the infants when their biological mothers are returned to prison. Pendleton and colleagues (2022) found that infants are most commonly discharged to grandparents and that transitions in care during the first year of the infant's life are common. Future research would benefit from a deeper understanding of how incarcerated mothers who give birth in custody arrange care for their infants and other children, caregivers' experiences providing (often temporary) care, how transitions in care are experienced by families over time, and children's developmental outcomes.

Notably, Kirubarajan's review (2022) only included studies published through December 2020 and therefore did not consider the considerable impacts that the COVID pandemic had on pregnant and birthing people in prisons and jails. A cross-sectional study found that prisons and jails made significant changes to the policies and practices related to prenatal care, programming, housing, and visiting for pregnant people (C. Kramer et al., 2022). Most notably, quarantine was commonly used as a COVID mitigation strategy, meaning that pregnant and postpartum people were routinely isolated upon their arrival to the prison, after they returned to the prison from off-site appointments, and even in the postpartum period following hospitalization for childbirth. Qualitative interviews with pregnant people incarcerated during the onset of COVID revealed that pregnant and postpartum people in prison experienced the pandemic as exacerbating already intolerable conditions (Marsh et al., 2024). Similarly, Abbott's (2023) in-depth interviews with women who provided pregnancy and birth support to incarcerated people revealed the complex ways in which prisons had to adapt services (e.g., virtual care and support) and the added risk that COVID had for pregnant patients' physical and mental health.

Policies related to pregnancy and incarceration

Throughout this paper we have highlighted several policy issues that constrain or enable adequate pregnancy and postpartum care for incarcerated individuals. There are further notable concerted policy efforts—and opportunities for improvement. Legislative policies are one strategy to address gaps in care for pregnant and postpartum individuals in custody. The most significant attention has

been directed against the routine practice of restraining pregnant individuals, especially when taken off-site for medical appointments or delivery. Between 1999 and 2024, 41 states, the District of Columbia, and the federal government passed laws prohibiting shackling of incarcerated people giving birth, with a surge of laws passed since 2014 in part due to activism of previously incarcerated women (Advocacy & Research on Reproductive Wellness of Incarcerated People, 2024). However, research and reports demonstrate that implementation and compliance are variable, and that shackling still happens even in states with laws. Some reasons for this are that all laws have exceptions, which hospital or custody staff may over-apply, and that hospital staff may be unaware of the laws or have stigmatizing attitudes (L. S. Goshin et al., 2019, 2020). Additionally, these laws do not apply to people in police custody. Some states' anti-shackling laws incorporate additional provisions, such as disallowing restraints at other points of pregnancy and postpartum, requiring access to doula support, or requiring written prenatal care policies.

Given the high prevalence of OUD in pregnancy among incarcerated individuals and the need to ensure timely access to lifesaving MOUD, legislation is one potential strategy to ensure access to this standard of care. A review of state laws through 2020 found that only seven states had laws related to SUD screening and treatment for pregnant individuals (but none specifying postpartum) in custody, with only three requiring access to MOUD (Steely Smith et al., 2023).

Although no state laws specifically address abortion access for incarcerated individuals, since the Supreme Court overturned *Roe v. Wade* in 2022 in *Dobbs v. Jackson Women's Health Organization* (*Dobbs*), at least 17 states have banned or severely restricted abortion care, including nine with no exception for rape (Institute, 2024). Incarcerated pregnant individuals in those states are not free to travel out of state, a prohibition that remains on parole or probation, making abortion inaccessible to them (Institute, 2024). Abortion bans have been shown to negatively impact care for life threatening pregnancy emergencies, increasing maternal morbidity and mortality. The effect of refusals of emergency abortion care on incarcerated people remains unexplored.

Despite noncompliance with existing laws and gaps in policy to improve the care of pregnant and postpartum people in custody, legislative action remains one way to make significant and noticeable impact on this population. There have been several federal efforts to codify improvements and enhancements to care for pregnant individuals in custody. For instance, the Justice for Incarcerated Moms Act and the Pregnant Women in Custody

Act have been introduced to Congress, though they have not been discussed and it is unclear whether they have a chance of being passed (*Booker, Durbin, Hirono Reinroduce Legislation to Address Maternal Health Crisis Among Incarcerated People | U.S. Senator Cory Booker of New Jersey, n.d.; Rep. Kamlager-Dove and Rep. Debbie Lesko Introduce Bipartisan Legislation to Protect Health of Newborns and Incarcerated Women, 2023*). The First Step Act of 2018, a sweeping criminal legal system reform law, included a small section that addressed banning the shackling of pregnant and laboring people who were in federal custody and required BJS to collect data on pregnancy outcomes in federal prisons (Sen. Sullivan, 2018). This strategy of incorporating pregnancy care in custody into other bills may be more successful in the short term, though incomplete in the long term. The Momnibus, a multiagency bill designed to decrease Black maternal mortality, for instance, includes provisions for incarcerated individuals (Sen. Booker, 2023). Another policy advancement is the inclusion of provisions for this population in appropriations bills. For instance, the 2021 U.S. House of Representatives Committee on Appropriations Report 116–455 directed BJS to conduct a feasibility study on collecting maternal health data from prisons and jails. BJS published the results of that feasibility study in 2024 (Irazola et al., 2024) and followed that up with a report in 2025 of pregnancy outcomes in state and federal prisons, the first ever federal report of its kind (Maruschak, 2025).

Together, these legislative actions help garner attention for pregnant and postpartum people in custody, an often-overlooked population but they lack implementation and enforcement provisions to make them successful. Relatedly, federal legislation does not address the variability in prison and jail operations, the lack of oversight on carceral operations, including health care, and the lack of sanctions for violations, nor does it remove the wide loopholes within state-level restraint laws. Lack of awareness of existing laws and insufficient accountability when anti-shackling laws are violated leads to the continued practice of restraining pregnant people in custody. Additionally, studies have documented that carceral facilities' policies may violate state law (C. Kramer et al., 2023) and that there is mismatch between jail (and state) policies and their reported practices (Kao et al., 2023) demonstrating that what happens on the ground is not always legal or guided by policy.

One potential avenue for federal policy reform is in the realm of Medicaid, which currently excludes incarcerated individuals as beneficiaries. Revising this eligibility to include incarcerated individuals could come with incentives for facilities to standardize and improve access to perinatal care, although it is unlikely to be a full solution

(Saloner et al., 2022). Several states have successively applied for 1115b waivers that allow incarcerated individuals to be covered by Medicaid for a designated amount of time before their release (Kaiser Family Foundation, 2025). Research is warranted to understand whether Medicaid expansion to incarcerated patients actually improves care and outcomes.

One promising legislative strategy recognizes the value of alternatives to incarceration for pregnant and parenting individuals. In 2021, Minnesota passed the Healthy Start Act, which diverts eligible pregnant individuals sentenced to state prison to the community for the duration of their pregnancy and 12 months postpartum (Healthy Start Act, 2021). This law resulted from the leadership of directly impacted people, and in partnership with researchers. Early evaluation findings, however, suggest that eligibility criteria are not well-defined and that few pregnant people admitted to Minnesota's state prison were released under this law (R. Shlafer & Osman, 2025). Several other states have introduced similar legislation, and Colorado implemented a similar law in 2023 (Alternatives In Criminal Justice System And Pregnant Persons, n.d. 2023). While many of these legislative successes have been informed by research, there is a dearth of research evaluating the implementation and impact of these laws. To advance meaningful change, research must accompany these initiatives.

Future directions

In summary, we have demonstrated that there is a widespread lack of systematic, national data about pregnancy and postpartum prevalence and outcomes for people in custody. Yet the connections to the maternal mortality crisis and broader issues of health inequities should not be overlooked and require deeper critical inquiry. Amid the data that does exist, we have shown that there are indeed pregnant individuals in custody, that some pregnancies end in custody, and that those pregnancies end in a variety of ways. There are numerous risk factors for adverse pregnancy outcomes and yet access to comprehensive pregnancy care is variable and often substandard. The complete lack of data on prevalence of postpartum individuals in custody, in the context of mental health conditions being both the leading cause of pregnancy-related deaths with more than half occurring postpartum and exceedingly common among incarcerated women should raise alarms—and signal a call to action for future research and policy.

Maternal health care services for incarcerated pregnant and postpartum people are essential for advancing public health, particularly given the distinct perinatal risks faced by this population. As researchers embark on the next decade of scholarship on this topic, we call for

centering the voices of directly impacted people in this effort. Research teams should actively engage people that are formerly incarcerated from idea generation to dissemination. Working with community research councils, advisory boards, community partners, and hiring people who have lived experience with the criminal legal system is critical to advancing health equity.

We recommend researchers utilize innovative approaches to answer complex questions related to the health of pregnant and postpartum incarcerated people. For example, longitudinal designs that consider dyadic and family-level measures will be important to understanding how incarceration impacts short and long-term maternal, child, and family health, including deepening our understanding of the harmful effects of separating infants from their mothers during incarceration. We further recommend greater attention to postpartum status, care, and outcomes for people who are postpartum in custody as well as their outcomes after they leave custody, thereby examining the impact of incarceration exposure on perinatal health, regardless of whether someone's pregnancy ends in custody. As reproductive health care services, abortion in particular, come under increasing and dynamic legislative and judicial regulation, it is imperative that research be done to investigate the ripple effect that such policies have on incarcerated pregnant people and their access to care.

Furthermore, there must be concerted efforts from carceral facilities, staff, and the Department of Justice to improve care for incarcerated pregnant and postpartum individuals. These efforts include knowledge building and improving awareness of existing laws and resources, standardizing care, technical assistance, and reporting maternal health data. In the 2024 U.S. GAO report, they found that very few state prisons and local jails received Health and Human Services and DOJ federal assistance despite grants being available to support maternal health care (G. Goodwin, 2024). Additionally, the National Institute of Corrections and BJA did not receive any requests for maternal health care technical assistance between 2018–2023. However, these opportunities may not exist for long under the 2025 presidential administration that has demonstrated efforts to making prisons and jails even worse and eliminating oversight and transparency in carceral facilities (*Tracking How the Trump Administration Is Making the Criminal Legal System Worse*, 2025). Notably, in response to administrative orders in 2025, BJS removed their questions related to gender identity in their surveys of people in jails, making it harder to understand gender-related experiences behind bars, including those related to pregnancy and women's health.

There is much to be learned from integrating existing data sources to answer new questions at the intersection

of maternal health and the criminal legal system. For example, McKenna and Nowotny (2024) recently integrated data from a variety of existing public sources to test whether mass incarceration is associated with infant mortality and low birthweight across U.S. counties. Leveraging existing population-based data will be an important avenue forward in the next decade but will not be enough to answer key questions. To that end, we recommend that researchers advocate for new and expanded measures of criminal legal system involvement to be added to existing public health surveillance systems (e.g., Pregnancy Risk Assessment Monitoring System [PRAMS]). Similarly, much could be learned if new measures related to reproductive health were added to existing surveys of state and federal prisons and jails (e.g., BJS's annual Survey of Prison Inmates).

Regardless of the data collected or the methods used, we implore researchers to prioritize the translation of their research to non-academic audiences, particularly practitioners and policymakers. As we have summarized here, there is a dearth of evidence related to this population; researchers must do a better job of recognizing the role of their research in informing the development and implementation of evidence-based practice and policy to advance health equity among pregnant and postpartum people impacted by the criminal legal system.

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References

- Abbott, L. (2023). *Pregnancy and new motherhood in prison during the COVID-19 pandemic*. Policy Press.
- ACOG. (2018). ACOG Committee Opinion No. 736: Optimizing Postpartum Care. *Obstetrics and Gynecology*, 131(5), e140–e150. <https://doi.org/10.1097/AOG.0000000000002633>
- Advocacy and Research on Reproductive Wellness of Incarcerated People. (2024). Pregnant While Incarcerated- Anti-Shackling Legislation and Resource Table. https://arrip.org/wp-content/uploads/2024/06/ARRWIP-Anti-shackling-legislation-and-resource-table_-Jan24.pdf
- Alexander M. (2010). *The New Jim Crow: Mass Incarceration in the Age of Colorblindness*. New York: New Press.
- Alternatives In Criminal Justice System And Pregnant Persons, HB23–1187, Colorado General Assembly 2023 Regular Session.
- American College of Obstetricians and Gynecologists. (2021). ACOG Practice Bulletin No. 830: Reproductive Health Care for Incarcerated Pregnant, Postpartum, and Nonpregnant Individuals. *Obstetrics and Gynecology*, 138(1), e24–e34. <https://doi.org/10.1097/AOG.00000000000004429>
- Asiodu, I. V., Beal, L., & Sufrin, C. (2021). Breastfeeding in Incarcerated Settings in the United States: A National Survey of Frequency and Policies. *Breastfeeding Medicine*, 16(9), 710–716. <https://doi.org/10.1089/bfm.2020.0410>
- Baker, B. (2019). Perinatal Outcomes of Incarcerated Pregnant Women: An Integrative Review. *Journal of Correctional Health Care*, 25(2), 92–104. <https://doi.org/10.1177/1078345819832366>
- Bard, E., Knight, M., & Plugge, E. (2016). Perinatal Health Care Services for Imprisoned Pregnant Women and Associated Outcomes: A Systematic Review. *BMC Pregnancy and Childbirth*, 16(1), 285. <https://doi.org/10.1186/s12884-016-1080-z>
- Barkauskas, V., Kane Low, L., & Pimlott, S. (2002). Health Outcomes of Incarcerated Pregnant Women and Their Infants in a Community-Based Program. *Journal of Midwifery and Women's Health*, 47(5), 371–379. [https://doi.org/10.1016/s1526-9523\(02\)00279-9](https://doi.org/10.1016/s1526-9523(02)00279-9)
- Beit, C. (2022). *Behind Bras and Bars: A Legal History of Breastfeeding in American Prisons*. Yale University.
- Bigelow, A., Power, M., MacLellan-Peters, J., Alex, M., & McDonald, C. (2012). Effect of mother/infant skin-to-skin contact on postpartum depressive symptoms and maternal physiological stress. *Journal of Obstetric, Gynecologic, and Neonatal Nursing: JOGNN*, 41(3), 369–382. <https://doi.org/10.1111/j.1552-6909.2012.01350.x>
- Binswanger, I. A., Merrill, J. O., Krueger, P. M., White, M. C., Booth, R. E., & Elmore, J. G. (2010). Gender Differences in Chronic Medical, Psychiatric, and Substance-Dependence Disorders Among Jail Inmates. *American Journal of Public Health*, 100(3), 476–482. <https://doi.org/10.2105/AJPH.2008.149591>
- Bohren, M. A., Hofmeyr, J. G., Sakala, C., Fukuzawa, R. K., Cuthbert, A., spsamps Cochrane Pregnancy and Childbirth Group. (2017). Continuous Support for Women during Childbirth. *Cochrane Database of Systematic Reviews*, 7. <https://doi.org/10.1002/14651858.CD003766.pub5>
- Booker, Durbin, Hirono Reintroduce Legislation to Address Maternal Health Crisis Among Incarcerated People | U.S. Senator Cory Booker of New Jersey. (n.d.). Retrieved September 9, 2024, from <https://www.booker.senate.gov/news/press/booker-durbin-hirono-reintroduce-legislation-to-address-maternal-health-crisis-among-incarcerated-people>
- Bradley, D. (2024). *Intimate Spaces in Carceral Places: An Examination of Black Doula Care with Incarcerated Pregnant and Postpartum Women*. Howard University.
- Bronson, J., Stroop, J., Zimmer, S., & Berzofsky, M. (2017). Drug Use, Dependence, and Abuse Among State Prisoners and Jail Inmates, 2007–2009. NCJ 250546. (NCJ250546). U.S. Department of Justice, Bureau of Justice Statistics; <https://bjs.ojp.gov/library/publications/drug-use-dependence-and-abuse-among-state-prisoners-and-jail-inmates-2007-2009>.
- Bronson, J., & Sufrin, C. (2019). Pregnant Women in Prison and Jail Don't Count: Data Gaps on Maternal Health and Incarceration. *Public Health Reports*, 134(1_suppl), 57S–62S. <https://doi.org/10.1177/0033354918812088>
- Bruzeli, E., & Martins, S. S. (2022). US Trends in Drug Overdose Mortality Among Pregnant and Postpartum Persons, 2017–2020. *JAMA*, 328(21), 2159–2161. <https://doi.org/10.1001/jama.2022.17045>
- Bureau of Justice Statistics. (2022). *Federal Prisoner Statistics Collected under the First Step Act, 2022*. Department of Justice, Bureau of Justice Statistics. <https://bjs.ojp.gov/library/publications/federal-prisoner-statistics-collected-under-first-step-act-2022>
- Byrne, M. W. (2019). Interventions in Prison Nurseries. In *Handbook on Children with Incarcerated Parents* (pp. 167–181). Springer.
- Care Post-Roe: How post-Roe laws are obstructing clinical care. (n.d.). ANSIRH. Retrieved September 16, 2024, from <https://www.ansirh.org/research/research/care-post-ro-how-post-ro-laws-are-obstructing-clinical-care>
- Carlson, J. R. (2001). Prison Nursery 2000: A Five-Year Review of the Prison Nursery at the Nebraska Correctional Center for Women. *Journal of Offender Rehabilitation*, 33(3), 75–97.
- Carlson, J. R., Jr. (2009). Prison Nurseries: A Pathway to Crime-Free Futures. *Correct Compend*, 34(1), 17–23.
- Carlson, J. R. (2018). Prison Nurseries: A Way to Reduce Recidivism. *The Prison Journal*, 98(6), 760–775. <https://doi.org/10.1177/0032885518812694>
- Carson, E. A. (2022). Prisoners in 2022. Department of Justice, Bureau of Justice Statistics. <https://bjs.ojp.gov/library/publications/prisoners-2022-statistical-tables>
- Carson, E. A., & Kluckow, R. (2023). Prisoners in 2022– Statistical Tables. (NCJ307149; Prisoners). U.S. Department of Justice, Bureau of Justice Statistics. <https://bjs.ojp.gov/library/publications/prisoners-2022-statistical-tables>
- Carter Ramirez, A., Liauw, J., Costescu, D., Holder, L., Lu, H., & Kouyoumdjian, F. G. (2020). Infant and Maternal Outcomes for Women Who Experience Imprisonment in Ontario, Canada: A Retrospective Cohort Study. *Journal of Obstetrics and Gynaecology Canada: JOGC = Journal d'obstetrique et Gynecologie Du Canada: JOGC*, 42(4), 462–472.e2. <https://doi.org/10.1016/j.jogc.2019.11.068>
- Cavanagh, A., Shamsheri, T., Shen, K., Gaber, J., Liauw, J., Vanstone, M., & Kouyoumdjian, F. (2022). Lived Experiences of Pregnancy and Prison through a Reproductive Justice Lens: A Qualitative Meta-Synthesis. *Social Science & Medicine* (1982), 307, 115179. <https://doi.org/10.1016/j.socscimed.2022.115179>
- CDC. (2024a, May 14). Breastfeeding Benefits Both Baby and Mom. Breastfeeding. <https://www.cdc.gov/breastfeeding/features/breastfeeding-benefits.html>
- CDC. (2024b, May 20). Pregnancy Mortality Surveillance System. Maternal Mortality Prevention. <https://www.cdc.gov/maternal-mortality/php/pregnancy-mortality-surveillance/index.html>
- Centers for Disease Control and Prevention. (2024, May 20). Pregnancy Mortality Surveillance System. Maternal Mortality Prevention. <https://www.cdc.gov/maternal-mortality/php/pregnancy-mortality-surveillance/index.html>
- Chambers, B. D., Arega, H. A., Arabia, S. E., Taylor, B., Barron, R. G., Gates, B., Scruggs-Leach, L., Scott, K. A., & McLemore, M. R. (2021). Black Women's Perspectives on Structural Racism across the Reproductive Lifespan: A Conceptual Framework for Measurement Development. *Maternal and Child Health Journal*, 25(3), 402–413. <https://doi.org/10.1007/s10995-020-03074-3>
- Chawla, A., Bansal, N. L., Liu, C., & Olagunju, A. T. (2024). Postpartum Depression in Correctional Populations. *Journal of Correctional Health Care*, 30(2), 65–70. <https://doi.org/10.1089/jchc.23.08.0071>
- Cohain, J. S., Buxbaum, R. E., & Mankuta, D. (2017). Spontaneous first trimester miscarriage rates per woman among parous women with 1 or more pregnancies of 24 weeks or more. *BMC Pregnancy and Childbirth*, 17, 437. <https://doi.org/10.1186/s12884-017-1620-1>
- Committee on Adolescence, Braverman, P. K., & Murray, P. J. (2011). Health Care for Youth in the Juvenile Justice System. *Pediatrics*, 128(6), 1219–1235. <https://doi.org/10.1542/peds.2011-1757>
- Craig, S. (2009). Historical Review of Mother and Child Programs for Incarcerated Women | Office of Justice Programs. *The Prison Journal*, 89(1), 35S–53S.
- Dallaire, D., Forestall, C., Kelsey, C., Ptachick, B., & MacDonnell, K. (2017). A nutrition-based program for pregnant incarcerated women. *Journal of Offender Rehabilitation*, 56(4), 277–294.
- Dehlendorf, C., Harris, L. H., & Weitz, T. A. (2013). Disparities in Abortion Rates: A Public Health Approach. *American Journal of Public Health*, 103(10), 1772–1779. <https://doi.org/10.2105/AJPH.2013.301339>
- DiNardo, C. (2018). Pregnancy in Confinement, Anti-Shackling Laws and the "Extraordinary Circumstances" Loophole. *Duke Journal of Gender Law & Policy*, 25(2), 271–295.

- Hinton, E. K. (2016). *From the War on Poverty to the War on Crime: the Making of mass incarceration in America*. Cambridge, Massachusetts: Harvard University Press.
- Faherty, L. J., Kranz, A. M., Russell-Fritch, J., Patrick, S. W., Cantor, J., & Stein, B. D. (2019). Association of Punitive and Reporting State Policies Related to Substance Use in Pregnancy With Rates of Neonatal Abstinence Syndrome. *JAMA Network Open*, 2(11), Article e1914078. <https://doi.org/10.1001/jamanetworkopen.2019.14078>
- Fair, H., & Walmsley, R. (2022). World Female Imprisonment List. Institute for Crime and Justice Policy Research. <https://www.icpr.org.uk/news-events/2022/world-female-prison-population-60-2000>
- Ferszt, G. G., & Clarke, J. G. (2012). Health Care of Pregnant Women in U.S. State Prisons. *Journal of Health Care for the Poor and Underserved*, 23(2), 557–569.
- Franco, C., Mowers, E., & Lewis, D. L. (2020). Equitable Care for Pregnant Incarcerated Women: Infant Contact After Birth-A Human Right. *Perspectives on Sexual and Reproductive Health*. <https://doi.org/10.1363/psrh.12166>
- Ghaedrahmati, M., Kazemi, A., Kheirabadi, G., Ebrahimi, A., & Bahrami, M. (2017). Postpartum depression risk factors: A narrative review. *Journal of Education and Health Promotion*, 6, 60. https://doi.org/10.4103/jehp.jehp_9_16
- Goodwin, G. (2024). Pregnant Women in State Prisons and Local Jails: Federal Assistance to Support Their Care (GAO-25–106404). United States Government Accountability Office. <https://www.gao.gov/products/gao-25-106404>
- Goodwin, M. (2020). *Policing the Womb: Invisible Women and the Criminalization of Motherhood*. Cambridge University Press.
- Goshin, L., Byrne, M., & Henninger, A. (2014). *Recidivism after Release from a Prison Nursery Program*, 31(2), 109–117.
- Goshin, L. S., Sissoko, D. R. G., Neumann, G., Sufrin, C., & Byrnes, L. (2019). Perinatal Nurses' Experiences With and Knowledge of the Care of Incarcerated Women During Pregnancy and the Postpartum Period. *Journal of Obstetric, Gynecologic, and Neonatal Nursing: JOGNN*, 48(1), 27–36. <https://doi.org/10.1016/j.jogn.2018.11.002>
- Goshin, L. S., Sissoko, D. R. G., Stringer, K. L., Sufrin, C., & Byrnes, L. (2020). Stigma and US Nurses' Intentions to Provide the Standard of Maternal Care to Incarcerated Women, 2017. *American Journal of Public Health*, 110(5), S93–S99. <https://doi.org/10.2105/AJPH.2019.305408>
- Hayes, C. M., Sufrin, C., & Peritt, J. B. (2020). Reproductive Justice Disrupted: Mass Incarceration as a Driver of Reproductive Oppression. *American Journal of Public Health*, 110(5), S21–S24. <https://doi.org/10.2105/AJPH.2019.305407>
- Healthy Start Act, SF 1315, State of Minnesota Senate Ninety-Second Session (2021). https://www.revisor.mn.gov/bills/text.php?number=SF1315&version=latest&session=ls92&session_year=2021&session_number=0
- Hendricks, C. A., Rajagopal, K. M., Sufrin, C. B., Kramer, C., & Jiménez, M. C. (2024). Mental health, chronic and infectious conditions among pregnant persons in US state prisons and local jails 2016–2017. *Women's Health*, 20, 17455057241228748. <https://doi.org/10.1177/17455057241228748>
- Hessami, K., Hutchinson-Colas, J. A., Chervenak, F. A., Shamshirsaz, A. A., Zargarzadeh, N., Norooznezhad, A. H., Grünebaum, A., & Bachmann, G. A. (2023). Prenatal care and pregnancy outcome among incarcerated pregnant individuals in the United States: A systematic review and meta-analysis. *Journal of Perinatal Medicine*, 51(5), 600–606. <https://doi.org/10.1515/jpm-2022-0412>
- Howard, D. L., Strobino, D., Sherman, S., & Crum, R. (2008). Within prisons, is there an association between the quantity of prenatal care and infant birthweight? *Paediatric and Perinatal Epidemiology*, 22(4), 369–378. <https://doi.org/10.1111/j.1365-3016.2008.00933.x>
- Howard, D. L., Strobino, D., Sherman, S. G., & Crum, R. M. (2011). Maternal Incarceration During Pregnancy and Infant Birthweight. *Maternal and Child Health Journal*, 15(4), 478–486. <https://doi.org/10.1007/s10995-010-0602-y>
- Howland, M. A., Kotlar, B., Davis, L., & Schlafer, R. J. (2021). Depressive Symptoms among Pregnant and Postpartum Women in Prison. *Journal of Midwifery & Women's Health*, 66(4), 494–502. <https://doi.org/10.1111/jmwh.13239>
- Huang, K., Atlas, R., & Parvez, F. (2012). The Significance of Breastfeeding to Incarcerated Pregnant Women: An Exploratory Study. *Birth*, 39(2), 145–155. <https://doi.org/10.1111/j.1523-536X.2012.00528.x>
- Insights into the U.S. Maternal Mortality Crisis: An International Comparison. (2024, June 4). <https://doi.org/10.26099/cthn-st75>
- Institute, G. (2024, September 25). Interactive Map: US Abortion Policies and Access After Roe. <https://states.guttmacher.org/policies/>
- Irazola, S., Maruschak, L., & Bronson, J. (2024). Data on Maternal Health and Pregnancy Outcomes from Prisons and Jails: Results from a Feasibility Study (NCJ 307326). Bureau of Justice Statistics. <https://bjs.ojp.gov/library/publications/data-maternal-health-and-pregnancy-outcomes-prisons-and-jails-results>
- Jones, R. K., Witwer, E., & Jerman, J. (2019). Abortion Incidence and Service Availability in the United States, 2017. <https://doi.org/10.1363/2019.30760>
- Kaiser Family Foundation. (2025, March 14). Medicaid Waiver Tracker: Approved and Pending Section 1115 Waivers by State. Medicaid Waiver Tracker. <https://www.kff.org/medicaid/issue-brief/medicaid-waiver-tracker-approved-and-pending-section-1115-waivers-by-state/>
- Kao, L., Lee, C., Parayil, T., Kramer, C., & Sufrin, C. B. (2023). Assessing provision of MOUD and obstetric care in U.S. jails: A content analysis of policies submitted by 59 jails. *Drug and Alcohol Dependence*, 248, 109877. <https://doi.org/10.1016/j.drugalcdep.2023.109877>
- Karvonen, K. L., McKenzie-Sampson, S., Baer, R. J., Jelliffe-Pawloski, L., Rogers, E. E., Pantell, M. S., & Chambers, B. D. (2023). Structural Racism is Associated With Adverse Postnatal Outcomes Among Black Preterm Infants. *Pediatric Research*, 94(1), 371–377. <https://doi.org/10.1038/s41390-022-02445-6>
- Kasdan, D. (2009). Abortion Access for Incarcerated Women: Are Correctional Health Practices in Conflict With Constitutional Standards. *Perspectives on Sexual and Reproductive Health*, 41(1), 59–62. <https://doi.org/10.1363/4105909>
- Kelsey, C. M., Medel, N., Mullins, C., Dallaire, D., & Forestell, C. (2017). An Examination of Care Practices of Pregnant Women Incarcerated in Jail Facilities in the United States. *Maternal and Child Health Journal*, 21(6), 1260–1266. <https://doi.org/10.1007/s10995-016-2224-5>
- Kim, M., Sufrin, C., Nowotny, K., Beal, L., & Jiménez, M. C. (2021). Pregnancy Prevalence and Outcomes in Three U.S. Juvenile Residential Systems. *Journal of Pediatric and Adolescent Gynecology*, 34(4), 546–551. <https://doi.org/10.1016/j.jpjag.2021.01.005>
- Kirksey, K. (2021). A social history of racial disparities in breastfeeding in the United States. *Social Science & Medicine*, 289, Article 114365. <https://doi.org/10.1016/j.socscimed.2021.114365>
- Kirubakaran, A., Tsang, J., Dong, S., Hui, J., Sreeram, P., Mohmand, Z., Leung, S., Ceccacci, A., & Sobel, M. (2022). Pregnancy and childbirth during incarceration: A qualitative systematic review of lived experiences. *BJOG: An International Journal of Obstetrics & Gynaecology*, 129(9), 1460–1472. <https://doi.org/10.1111/1471-0528.17137>
- Knight, M., & Plugge, E. (2005). Risk Factors for Adverse Perinatal Outcomes in Imprisoned Pregnant Women: A Systematic Review. *BMC Public Health*, 5, 111. <https://doi.org/10.1186/1471-2458-5-111>
- Knittel, A. K. (2019). "Resolving Health Disparities for Women Involved in the Criminal Justice System." *North Carolina Medical Journal* 80(6): 363–66. <https://doi.org/10.18043/ncm.80.6.363>
- Knittel, A. K., Ferguson, E. G., Balasubramanian, V., Carda-Auten, J., DiRosa, E., & Rosen, D. L. (2023). "We don't wanna birth it here": A qualitative study of Southern jail personnel approaches to pregnancy. *Women & Criminal Justice*, 33(5), 349–362. <https://doi.org/10.1080/08974454.2022.2040693>
- Knittel, A., & Sufrin, C. (2020). Maternal Health Equity and Justice for Pregnant Women Who Experience Incarceration. *JAMA Network Open*, 3(8), e2013096. <https://doi.org/10.1001/jamanetworkopen.2020.13096>
- Knittel, A. K., Zarnick, S., Thorp, J. M., Amos, E., & Jones, H. E. (2020). Medications for opioid use disorder in pregnancy in a state women's prison facility. *Drug and Alcohol Dependence*, 214, Article 108159. <https://doi.org/10.1016/j.drugalcdep.2020.108159>
- Kortsmit, K. (2023). Abortion Surveillance—United States, 2021. *MMWR. Surveillance Summaries*, 72. <https://doi.org/10.15585/mmwr.ss7209a1>
- Kramer, C. T., Thomas, K., Hayes, C. M., Su, V., & Sufrin, C. B. (2025). "I'm giving birth in thirty days, and I still know nothing": A Qualitative Analysis of Incarcerated Pregnant People's Perceptions while considering a Prison Nursery Program. *Women & Criminal Justice*, 1–18. <https://doi.org/10.1080/08974454.2025.2492321>
- Kramer, C., Thomas, K., Patil, A., Hayes, C. M., & Sufrin, C. B. (2023). Shackling and pregnancy care policies in US prisons and jails. *Maternal*

- and Child Health Journal, 27(1), 186–196. <https://doi.org/10.1007/s10995-022-03526-y>
- Kramer, C., Williamston, A.-D., Shlafer, R. J., & Sufrin, C. B. (2022). COVID-19's Effect on Pregnancy Care for Incarcerated People. *Health Equity*, 6(1), 406–411. <https://doi.org/10.1089/hecq.2022.0035>
- Kwarteng-Amaning, V., Svoboda, J., Bachynsky, N., & Linthicum, L. (2019). An Alternative to Mother and Infants Behind Bars: How One Prison Nursery Program Impacted Attachment and Nurturing for Mothers Who Gave Birth While Incarcerated. *The Journal of Perinatal & Neonatal Nursing*, 33(2), 116. <https://doi.org/10.1097/JPN.0000000000000398>
- Labor Induction. (n.d.). Retrieved October 8, 2024, from <https://www.acog.org/womens-health/faqs/labor-induction>
- Lingerfelt, C., Hutson, S., Thomas, S., & Morgan, K. H. (2024). An Interpretive Description of Drug Withdrawal Among Pregnant Women in Jail. *Nursing for Women's Health*, 28(3), 187–198. <https://doi.org/10.1016/j.nwh.2023.12.002>
- Lipnicky, A., Stites, S., Sufrin, C., Bello, J. K., Shlafer, R., Kelly, P. J., & Ramaswamy, M. (2023). Jail Provision of Pregnancy and Sexual Health Services in Four Midwestern States. *Women's Health Issues*, 33(1), 97–104. <https://doi.org/10.1016/j.whi.2022.07.004>
- Logue, T. C., Wen, T., Staniczenko, A., Huang, Y., D'Alton, M. E., & Friedman, A. M. (2022). Delivery Hospitalizations among Incarcerated Women. *American Journal of Obstetrics and Gynecology*, S0002–9378(22), 00259–00269. <https://doi.org/10.1016/j.ajog.2022.03.057>
- Loretta Ross & Rickie Solinger. (2017). *Reproductive Justice: A New Vision for the 21st Century*. University of California Press. <http://www.ucpress.edu/book.php?isbn=9780520288201>
- Lynch, S. M., Fritch, A., & Heath, N. M. (2012). Looking Beneath the Surface: The Nature of Incarcerated Women's Experiences of Interpersonal Violence, Treatment Needs, and Mental Health. *Feminist Criminology*, 7(4), 381–400. <https://doi.org/10.1177/1557085112439224>
- Marsh, L. M., Kramer, C. T., Shlafer, R. J., & Sufrin, C. B. (2024). Impacts of the COVID-19 Pandemic on the Experiences of Incarcerated Pregnant People. *Health & Justice*, 12(40). <https://doi.org/10.1186/s40352-024-00296-3>
- Maruschak, L. (2008). Medical Problems of Prisoners | Bureau of Justice Statistics (Bureau of Justice Statistics Reports NCJ 221740). Department of Justice. <https://bjs.ojp.gov/library/publications/medical-problems-prisoners>
- Maruschak, L. (2025). Maternal Healthcare and Pregnancy Prevalence and Outcomes in Prisons, 2023 | Bureau of Justice Statistics. Department of Justice, Bureau of Justice Statistics. <https://bjs.ojp.gov/library/publications/maternal-healthcare-and-pregnancy-prevalence-and-outcomes-prisons-2023>
- Maruschak, L., & Beck, A. (2001). Medical Problems of Inmates, 1997 | Bureau of Justice Statistics (Medical Problems of Inmates). Department of Justice. <https://bjs.ojp.gov/library/publications/medical-problems-inmates-1997>
- Maruschak, L., Bronson, J., & Alper, M. (2021a). Indicators of Mental Health Problems Reported by Prisoners: Survey of Prison Inmates, 2016 (252643; Indicators of Mental Health Problems Reported by Prisoners). Department of Justice. <https://bjs.ojp.gov/library/publications/indicators-mental-health-problems-reported-prisoners-survey-prison-inmates>
- Maruschak, L., Bronson, J., & Alper, M. (2021b). Medical Problems Reported by Prisoners: Survey of Prison Inmates, 2016 | Bureau of Justice Statistics (NCJ 252644; Medical Problems Reported by Prisoners). Department of Justice. <https://bjs.ojp.gov/library/publications/medical-problems-reported-prisoners-survey-prison-inmates-2016>
- McKenna, M., & Nowotny, K. M. (2024). Mass Incarceration, Maternal Vulnerability, and Birth Outcomes Across U.S. Counties. *Maternal and Child Health Journal*. <https://doi.org/10.1007/s10995-024-03960-0>
- Meine, K. (2018). Pregnancy unshackled: Increasing equity through implementation of perinatal depression screening, shared decision making, and treatment for incarcerated women. *Nursing Forum*, 53(4), 437–447. <https://doi.org/10.1111/nuf.12271>
- Mental Health Treatment While Incarcerated. (n.d.). NAMI. Retrieved September 13, 2024, from <https://www.nami.org/advocacy/policy-priorities/improving-health/mental-health-treatment-while-incarcerated/>
- More Imprisonment Does Not Reduce State Drug Problems. (2018, March 8). Pew Trusts. <http://pew.org/2tszeZl>
- Moss, A., Pesqueira, C., Thomas, K., Benning, S., Shlafer, R., Kramer, C., & Sufrin, C. (2024, August). Reproductive Health Care Legislation for Incarcerated People. Advocacy and Research on Reproductive Wellness of Incarcerated People, Johns Hopkins University School of Medicine. https://arrwip.org/wp-content/uploads/2024/08/ARRWIP-Reproductive-Health-Laws-Table_Aug-2024-1.pdf
- Murphy, S. (1999). *1949- Combating stereotypes and stigma*. Rutgers University Press.
- Nielsen, T., Bernson, D., Terplan, M., Wakeman, S. E., Yule, A. M., Mehta, P. K., Bharel, M., Diop, H., Taveras, E. M., Wilens, T. E., & Schiff, D. M. (2020). Maternal and infant characteristics associated with maternal opioid overdose in the year following delivery. *Addiction*, 115(2), 291–301. <https://doi.org/10.1111/add.14825>
- Opposition to Criminalization of Individuals During Pregnancy and the Postpartum Period. (n.d.). Retrieved September 13, 2024, from <https://www.acog.org/clinical-information/policy-and-position-statements/state-ments-of-policy/2020/opposition-criminalization-of-individuals-pregnancy-and-postpartum-period>
- Pendleton, V. E., Schmitgen, E. M., Davis, L., & Shlafer, R. J. (2022). Caregiving Arrangements and Caregiver Well-being when Infants are Born to Mothers in Prison. *Journal of Child and Family Studies*, 31(7), 1894–1907. <https://doi.org/10.1007/s10826-021-02089-w>
- Pfaff, J. (2017). *Locked In: The True Causes of Mass Incarceration and How to Achieve Real Reform*. Basic Books.
- Ranji, U., Diep, K., & Published, A. S. (2024, June 21). Key Facts on Abortion in the United States. KFF. <https://www.kff.org/other/issue-brief/key-facts-on-abortion-in-the-united-states/>
- Rep. Kamlager-Dove and Rep. Debbie Lesko Introduce Bipartisan Legislation to Protect Health of Newborns and Incarcerated Women. (2023, February 21). Congresswoman Sydney Kamlager-Dove. <http://kamlager-dove.house.gov/media/press-releases/rep-kamlager-dove-and-rep-debbie-lesko-introduce-bipartisan-legislation>
- Roberts, D. (1997). *Killing the Black Body: Race, Reproduction, and the Meaning of Liberty*. Pantheon Books.
- Roberts, D. E. (2022). *Torn Apart: How the Child Welfare System Destroys Black Families—and how Abolition Can Build a Safer World (First edition)*. Basic Books.
- Reproductive Health Care for Incarcerated Pregnant, Postpartum, and Non-pregnant Individuals: ACOG Committee Opinion, Number 830. (2021). *Obstetrics & Gynecology*, 138(1), e24. <https://doi.org/10.1097/AOG.0000000000004429>
- Saloner, B., Eber, G. B., Sufrin, C. B., Beyrer, C., & Rubenstein, L. S. (2022). A Human Rights Framework for Advancing the Standard of Medical Care for Incarcerated People in the United States in the Time of COVID-19. *Health and Human Rights*, 24(1), 59–75.
- Schroeder, C., & Bell, J. (2005). Doula Birth Support for Incarcerated Pregnant Women. *Public Health Nursing*, 22(1), 53–58. <https://doi.org/10.1111/j.0737-1209.2005.22108.x>
- Sen. Booker, C. A. [D-N. (2023, May 15)]. S.1606 - 118th Congress (2023–2024): Black Maternal Health Momnibus Act (2023–05–15) [Legislation]. <https://www.congress.gov/bills/118th-congress/senate-bill/1606>
- Sen. Sullivan, D. [R-A. (2018, December 21)]. S.756 - 115th Congress (2017–2018): First Step Act of 2018 (12/21/2018) [Legislation]. <http://www.congress.gov/>
- Shlafer, R. J., Davis, L., Hindt, L. A., Goshin, L. S., & Gerrity, E. (2018). Intention and Initiation of Breastfeeding Among Women Who Are Incarcerated. *Nursing for Women's Health*, 22(1), 64–78. <https://doi.org/10.1016/j.nwh.2017.12.004>
- Shlafer, R., Davis, L., Hindt, L., & Pendleton, V. (2021). The Benefits of Doula Support for Women Who Are Pregnant in Prison and Their Newborns. In Children with Incarcerated Mothers: Separation, Loss, and Reunification (pp. 33–48). Springer.
- Shlafer, R. J., Hardeman, R. R., & Carlson, E. A. (2019). Reproductive justice for incarcerated mothers and advocacy for their infants and young children. *Infant Mental Health Journal*, 40(5), 725–741. <https://doi.org/10.1002/imhj.21810>
- Shlafer, R. J., Hellerstedt, W. L., Secor-Turner, M., Gerrity, E., & Baker, R. (2015). Doulas' Perspectives about Providing Support to Incarcerated Women: A Feasibility Study. *Public Health Nursing* (Boston, Mass.), 32(4), 316–326. <https://doi.org/10.1111/phn.12137>

- Shlafer, R., & Osman, I. (2025). *Implementation of Minnesota's Healthy Start Act*. University of Minnesota.
- Shlafer, R., Stang, J., Dallaire, D., Forestall, C. A., & Hellerstedt, W. (2017). Best Practices for Nutrition Care of Pregnant Women in Prison. *Journal of Correctional Healthcare*, 23(3), 297–304.
- State Bans on Abortion Throughout Pregnancy | Guttmacher Institute. (2024, May 1). <https://www.guttmacher.org/state-policy/explore/state-policies-abortion-bans>
- Steely Smith, M. K., Hinton-Froese, K. E., Scarbrough Kamath, B., Virmani, M., Walters, A., & Zielinski, M. J. (2024). Characteristics and Outcomes of Women and Infants Who Received Prenatal Care While Incarcerated in Arkansas State Prison System, 2014–2019. *Maternal and Child Health Journal*. <https://doi.org/10.1007/s10995-023-03875-2>
- Steely Smith, M. K., Zielinski, M. J., Sufrin, C., Kramer, C. T., Benning, S. J., Laine, R., & Shlafer, R. J. (2023). State Laws on Substance Use Treatment for Incarcerated Pregnant and Postpartum People. *Substance Abuse: Research and Treatment*, 17, 11782218231195556. <https://doi.org/10.1177/11782218231195556>
- Sufrin, C. (2017). Jailcare: Finding the Safety Net for Women Behind Bars. Oakland: California.
- Sufrin, C., Beal, L., Clarke, J., Jones, R., & Mosher, W. D. (2019). Pregnancy Outcomes in US Prisons, 2016–2017. *American Journal of Public Health*, 109(5), 799–805. <https://doi.org/10.2105/AJPH.2019.305006>
- Sufrin, C. B., Devon-Williamson, A., Beal, L., Hayes, C. M., & Kramer, C. (2023). "I mean, I didn't really have a choice of anything: "How incarceration influences abortion decision-making and precludes access in the United States. *Perspectives on Sexual and Reproductive Health*, 55(3), 165–177. <https://doi.org/10.1363/psrh.12235>
- Sufrin, C., Jones, R. K., Beal, L., Mosher, W. D., & Bell, S. (2021). Abortion Access for Incarcerated People: Incidence of Abortion and Policies at U.S. Prisons and Jails. *Obstetrics & Gynecology*, 138(3), 330. <https://doi.org/10.1097/AOG.0000000000004497>
- Sufrin, C., Jones, R. K., Mosher, W. D., & Beal, L. (2020). Pregnancy Prevalence and Outcomes in U.S. Jails. *Obstetrics and Gynecology*, 135(5), 1177–1183. <https://doi.org/10.1097/AOG.0000000000003834>
- Sufrin, C., Kolbi-Molinas, A. (2015). Roth R. Reproductive Justice, Health Disparities, and Incarcerated Women in the United States. *Perspectives in Sexual and Reproductive Health*, 47(4), 213–9.
- Sufrin, C., Kramer, C. T., Terplan, M., Fiscella, K., Olson, S., Voegtline, K., & Latkin, C. (2022). Availability of Medications for the Treatment of Opioid Use Disorder Among Pregnant and Postpartum Individuals in US Jails. *JAMA Network Open*, 5(1), Article e2144369. <https://doi.org/10.1001/jamanetworkopen.2021.44369>
- Sufrin, C., Sutherland L., Beal L., Terplan M., Latkin C., & Clarke J. G. (2020). Opioid use disorder incidence and treatment among incarcerated pregnant women in the United States: Results from a national surveillance study. *Addiction (Abingdon, England)*, 115(11), 2057–2065. <https://doi.org/10.1111/add.15030>
- Sufrin, C., Ziv, T., Dayton, L., Latkin, C., & Kramer, C. (2024). "They talked to me like I was dirt under their feet:"Treatment and withdrawal experiences of incarcerated pregnant people with opioid use disorder in four U.S. states. *SSM - Qualitative Research in Health*, 6, 100453. <https://doi.org/10.1016/j.ssmqr.2024.100453>
- Testa, A., Fahmy, C., Jackson, D. B., Ganson, K. T., & Nagata, J. M. (2022). Incarceration Exposure During Pregnancy and Maternal Disability: Findings from the Pregnancy Risk Assessment Monitoring System. *BMC Public Health*, 22(1), 744. <https://doi.org/10.1186/s12889-022-13143-7>
- The Turnaway Study. (n.d.). ANSIRH. Retrieved September 16, 2024, from <https://www.ansirh.org/research/ongoing/turnaway-study>
- Tracking how the Trump administration is making the criminal legal system worse. (2025, April 30). Prison Policy Initiative. <https://www.prisonpolicy.org/federaltracker.html>
- Trost, S., Beauregard, J., Chandra, G., Njie, F., Berry, J., Harvey, J., & Goodman, D. A. (2022). Pregnancy-Related Deaths: Data from Maternal Mortality Review Committees in 36 US States, 2017–2019 | CDC. Centers for Disease Control and Prevention. https://www.cdc.gov/maternal-mortality/php/data-research/mmrc-2017-2019.html?CDC_AAref_Val=https://www.cdc.gov/reproductivehealth/maternal-mortality/erase-mm/data-mmrc.html
- Tuxhorn, R. (2022). "I've Got Something to Live for Now": A Study of Prison Nursery Mothers. *Critical Criminology*, 30, 421–441. <https://doi.org/10.1007/s10612-020-09545-x>
- United States Department of Justice. (n.d.). Prison Rape Elimination Act (PREA) Standard 115.83: Ongoing medical and mental health care for sexual abuse victims and abusers. Retrieved March 31, 2025, from <https://www.prearesourcecenter.org/standard/115-83>
- Understanding Second Trimester Miscarriage | Family Planning | Obstetrics and Gynecology | UC Davis Health. (n.d.). Retrieved September 16, 2024, from <https://health.ucdavis.edu/obgyn/specialties/family-planning/early-pregnancy-miscarriage/second-trimester-miscarriage>
- Volkow, N. D. (2021). Addiction Should Be Treated, Not Penalized. *American College of Neuropsychopharmacology*, 46, 2048–2050.
- Weber, A., Miskle, B., Lynch, A., Arndt, S., & Acion, L. (2021). Substance Use in Pregnancy: Identifying Stigma and Improving Care. *Substance Abuse and Rehabilitation*, 12, 105–121. <https://doi.org/10.2147/SAR.S319180>
- Wennerstrom, A., Sugarman, M., Martin, D., Lobre, C. B., Haywood, C. G., & Niyogi, A. (2022). 'You have to be almost dead before they ever really work on you in prison': A qualitative study of formerly incarcerated women's health care experiences during incarceration in Louisiana, U.S. *Health & Social Care in the Community*, 30(5), 1763–1774. <https://doi.org/10.1111/hsc.13556>
- Zeng, Z. (2023). Jail Inmates in 2022– Statistical Tables. (NCJ307086; Jail Inmates, Prison and Jail Inmates at Midyear). Bureau of Justice Statistics. <https://bjs.ojp.gov/library/publications/jail-inmates-2022-statistical-tables>

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